Project Title	**Rising Waters: Machine Learning Based Flood Prediction System**		
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1. Brainstorming & Idea Prioritization

As this is an individually developed project, brainstorming was carried out as a solo ideation exercise rather than a group workshop. The exercise began by mapping the broader disaster-preparedness domain, then narrowing toward a problem that could realistically be solved end-to-end by a single developer within the available timeline, using a dataset that was actually obtainable.

The starting point was a simple observation: flood forecasting in most regions is handled at a broad, regional level by government meteorological departments, while the individuals and local bodies who need to act on that information are often left with generic bulletins rather than a quick, personalized risk read-out. This gap between “data exists” and “data is usable by an ordinary person” became the anchor for every idea considered below.

1.1 Problem Space Exploration

- Flood-prone regions often rely on delayed or generalized government advisories rather than instant, localized risk feedback.
- Historical meteorological data (rainfall, humidity, cloud cover) is available but not translated into an accessible predictive tool for residents or local authorities.
- A lightweight, data-driven web tool could offer instant flood-risk feedback without requiring specialized hardware.
- Existing weather apps report raw measurements (mm of rainfall, % humidity) without translating them into a risk judgment or a recommended action, leaving the interpretation burden on the user.
- A solo developer has limited time and no field team, so any chosen idea needed to work from a static, downloadable dataset rather than live sensor feeds or partnerships with external agencies.

1.2 Idea Prioritization Matrix

Each candidate idea below was scored informally on feasibility (achievable within a solo timeline with available data/skills) and impact (how directly it addresses the core problem), which together produced a rough priority ranking.

Idea	Feasibility	Impact	Priority
ML-based flood risk prediction from historical rainfall/weather data	High	High	P0 — Selected
Rule-based static threshold alert system (no ML)	High	Low	P3
IoT sensor-based real-time river/water-level monitoring	Low	High	P2 — Future Scope
Satellite-imagery based flood extent mapping	Low	Medium	P3
Chatbot-based citizen flood advisory assistant	Medium	Medium	P2 — Future Scope

1.3 Selected Idea & Rationale

The selected idea is a Machine Learning classification system trained on historical meteorological and rainfall data (temperature, humidity, cloud cover, seasonal and annual rainfall figures), served through a lightweight Flask web application. This was prioritized for its high feasibility as a solo build, direct use of an available structured dataset, and ability to deliver an immediately usable prediction interface without external hardware or API dependencies.

A second consideration behind this choice was the ability to demonstrate rigor without over-engineering: comparing multiple classical ML algorithms (rather than committing to a single model upfront) allows the final system to justify its model choice with evidence, while still remaining small enough for one person to build, train, and serve within the project timeline. IoT and satellitebased ideas were explicitly deferred to the future-scope backlog (see Section 20) rather than discarded, since they remain natural extensions once real-time data sources are available.